Eosense Flux Chambers: Background and Theory

Emily Chua

April 27, 2026

1 Original eosFD Theoretical Framework

1.1 eosFD Overview

The eosFD (Eosense Inc.) is a flux chamber system designed for high temporal-resolution monitoring of near-surface soil CO_2 fluxes. An eosFD Network consists of paired "Soil" (hereafter called "Sample") and "Reference" Nodes that measure gas concentrations within enclosed chambers and use their difference to estimate a steady-state CO_2 flux (Table 1).

The eosFD operates using a forced diffusion approach (Risk et al., 2011), in which gas exchange between the chamber and its surroundings is regulated by hydrophobic membranes with known conductance. Each Node contains side-wall membranes that permit exchange with the atmosphere, while the Sample Node additionally includes a bottom membrane that allows exchange with the underlying surface. The Reference Node is capped at the base, preventing exchange and providing a real-time atmospheric correction.

An equilibrium is established between gas entering the chamber through the bottom membrane and gas exiting through the side membranes. Measurements of gas concentration within each chamber are then used, together with known geometry and calibrated membrane properties, to estimate surface CO_2 flux.

1.2 Concentration Measurement

Gas-phase CO_2 concentrations within each eosFD chamber are measured using a non-dispersive infrared (NDIR) sensor housed within each Node (Fig. 1). The sensor measures CO_2 gas concentration along with internal temperature and pressure, allowing concentrations to be expressed on a molar basis.

Gas diffuses through the chamber membranes and equilibrates within the chamber volume, where it is sampled continuously by the NDIR sensor. Like all optical gas sensors, NDIR sensors exhibit gradual drift over time due to changes in their electrical and mechanical properties. To account for this drift, Eosense recommends users periodically check calibration of the sensor using known concentration standards, including a zero-CO_2 gas (e.g., high-purity N_2) and a span gas ($\sim$1,000 ppm CO_2).

Eosense calibrates concentration within the 0–30,000 ppm range, with an accuracy of $\pm$40 ppm within the 0–3,000 ppm range (Table 2). These measured gas-phase concentrations form the primary observables used in the flux calculations described below.

1.3 Flux Estimation

Flux across the membranes is driven by the concentration on either side and by the membrane's physical dimensions and diffusivity. The full differential equation is:

$$\frac{V_m}{S}\frac{\partial C_s}{\partial t} = f_s(t) - D\frac{C_s - C_r}{L} \tag{1}$$

where V_m is the volume of the measuring chamber, S is the surface area of the membrane, $\frac{\partial C_s}{\partial t}$ is the time rate of change of concentration in the chamber, $f_s(t)$ is the time-dependent flux rate from the sample, C_s is the concentration in the Sample chamber, C_r is the ambient concentration measured by the Reference chamber, D is the diffusivity through the chamber volume and across the membrane to the atmosphere, and L is the characteristic path length for gas diffusion through the membranes and chamber to the NDIR sensor (Fig. 1).

Eosense flux estimates assume that the change in the head space concentration reaches steady state over the time span of the concentration measurements (~10 s):

$$\frac{\partial C_s}{\partial t} = 0$$

This steady-state assumption reduces the differential equation to a linear dependence. Because the diffusivity and path length are constant, they can be represented by a single coefficient, $G = D/L$, which can be empirically measured using a known flux source (see Sect. 1.4).

$$f_s = G(C_s - C_r) \tag{2}$$

Flux is calibrated within 0–10 μmol m^{-2} s^{-1} range with a resolution of 0.2 μmol m^{-2} s^{-1} (Table 2).

1.4 Eosense's Calibration Method

Eosense calibrates for G by using a flux generator where F_s is known and rearranging Eq. 2. While it is possible to measure D, it differs slightly between different batches of membranes. It is also possible to estimate L, but obstructions in the flowpath from the internal CO_2 sensor and complex geometries (side vs. bottom membranes) make it difficult. Therefore, Eosense deems it easiest to empirically measure the characteristic G of each chamber.

The average calibration coefficient provided by Eosense is $G = 0.0123$ μmol m^{-2} s^{-1} ppm^{-1} (or since 1 ppm = 1 μmol mol^{-1}, units of mol m^{-2} s^{-1}). From the ideal gas law for CO_2 at 25^{o}C:

$$V = \frac{RT}{P} = \frac{(8.314\,\mathrm{kPa\,L\,mol^{-1}\,K^{-1}})(298.15\,\mathrm{K})}{101.325\,\mathrm{kPa}} = 0.024\,46\,\mathrm{m^3\,mol^{-1}}$$

We can then convert G to units of m s^{-1}:

$$0.0123\,\mathrm{mol\,m^{-2}\,s^{-1}} \times 0.024\,46\,\mathrm{m^3\,mol^{-1}} = \boxed{3.01 \times 10^{-4}\,\mathrm{m\,s^{-1}}}$$

Using an estimated path length of $L = 0.0318$ m (Table 4), the membrane diffusivity is estimated as $D = G \cdot L = 0.957 \times 10^{-5}\,\mathrm{m^2\,s^{-1}}$.

Table 1: Eosense terminology for an eosFD network, capable of high temporal resolution monitoring of surface CO_2 efflux.

Term	Description
Sample Node	A node configured to measure gas emissions from the sampling surface (soil, water, or other substrate). Its bottom membrane is uncapped, allowing diffusive gas exchange with the measurement surface.
Reference Node	A node configured to measure background (ambient) atmospheric gas concentrations. Its bottom membrane is capped, preventing gas exchange with the sampling surface.
Pair or Network	A matched set consisting of one Sample Node and one Reference Node. Together, the differential concentration between the nodes can be used to calculate gas flux.
Measuring chamber	A node's enclosed chamber in which gas concentration is measured. The membranes on the nodes provide a known and quantifiable gas conductance, restricting gas flow out of the chamber and enabling flux to be calculated from the rate of concentration change within the chamber over time.
Collar	A 2.5 cm-long attachment fitted to the chamber that forms a gas-tight seal with the measurement surface.

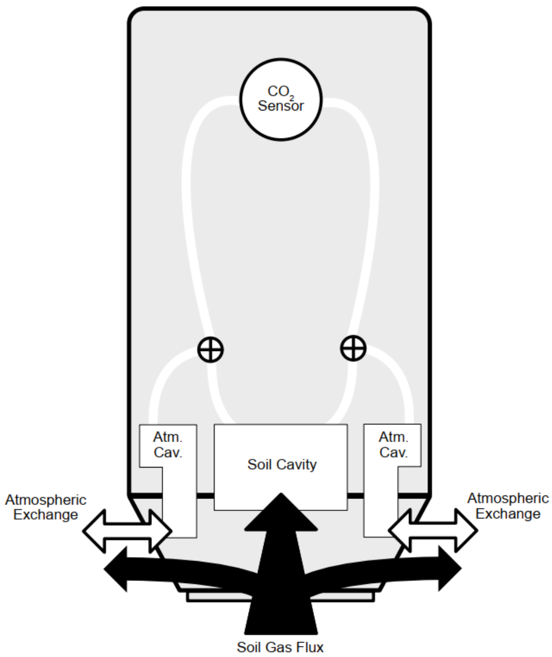

Figure 1: Internal view of the eosFD chamber, depicting the exchange of atmospheric gases and soil (sample) flux with the chamber cavities. Source: Nickerson et al. (2018).

Table 2: Performance specification details for the eosFDs, as provided by the Eosense manual.

Spec	**Value**
Calibration range (concentration)	0–30,000 ppm
Calibration range (flux)	0–10 μmol m^{-2} s^{-1}
Concentration accuracy	0–3,000 ± 40 ppm
	3,000–10,000 ± 2% reading
	Up to 30,000 ± 3.5% reading
Flux resolution	0.2 μmol m^{-2} s^{-1}

2 Reconsideration of eosFD Theory for Air–Sea Gas Flux Measurements

While Sect. 1 summarizes the theoretical framework and equations as provided by Eosense, this section revisits these assumptions and reformulates the eosFD mass balance to reflect its operation when deployed over an air–water interface.

Table 3: Nomenclature of the relevant parameters for ultimately calculating the air–water CO_2 flux rate beneath the eosFD chamber.

Param.	**Description**	**Units**	**Type**
V_m	Volume of measuring chamber	m^3	Known
V_c	Volume of collar	m^3	Calculated
S_a	Total surface area of side membranes	m^2	Known
S_c	Surface area of bottom membrane	m^2	Known
S_w	Surface area of water covered by collar	m^2	Known
C_a	Atmospheric CO_2 concentration	ppm	Measured
C_w	Waterside CO_2 concentration	ppm	Measured
C_s	CO_2 concentration in Sample Node chamber	ppm	Measured
C_r	CO_2 concentration in Reference Node chamber	ppm	Measured
C_c	CO_2 concentration in collar	ppm	Computed
k_a	Gas transfer velocity of side membrane	m s^{-1}	Calibrated
k_c	Gas transfer velocity of bottom membrane	m s^{-1}	Calibrated
k_w	Gas transfer velocity of water within collar	m s^{-1}	Computed
$k_w^\dagger$	"True" gas transfer velocity of water (no chamber)	m s^{-1}	Calibrated
f_a	CO_2 flux through Sample Node side membrane	mol m^{-2} s^{-1}	Intermediate
$f_{a,r}$	CO_2 flux through Reference Node side membrane	mol m^{-2} s^{-1}	Intermediate
f_c	CO_2 flux through bottom membrane	mol m^{-2} s^{-1}	Computed
f_w	CO_2 flux from water beneath chamber	mol m^{-2} s^{-1}	Computed
$f_w^\dagger$	"True" CO_2 flux from water (no chamber)	mol m^{-2} s^{-1}	Computed

Table 4: Eosense-provided values for relevant parameters in the flux calculations.

Param.	Description	Value	Source
G	Average membrane gas transfer velocity	3.01 x 10^{-4} m s^{-1}	Calibration note
L	Estimated diffusion path length	0.0318 m	Calibration note
S_a	Total surface area of side membranes	804 mm^2	Manual
S_c	Surface area of bottom membrane	804 mm^2	Manual
V_m	Volume of measuring chamber	16.7 cm^3	Michelle Coleman's email (4/16/25)
V_c	Volume of collar	Diameter = 5 cm; length depends on water height	Dimensions from manual

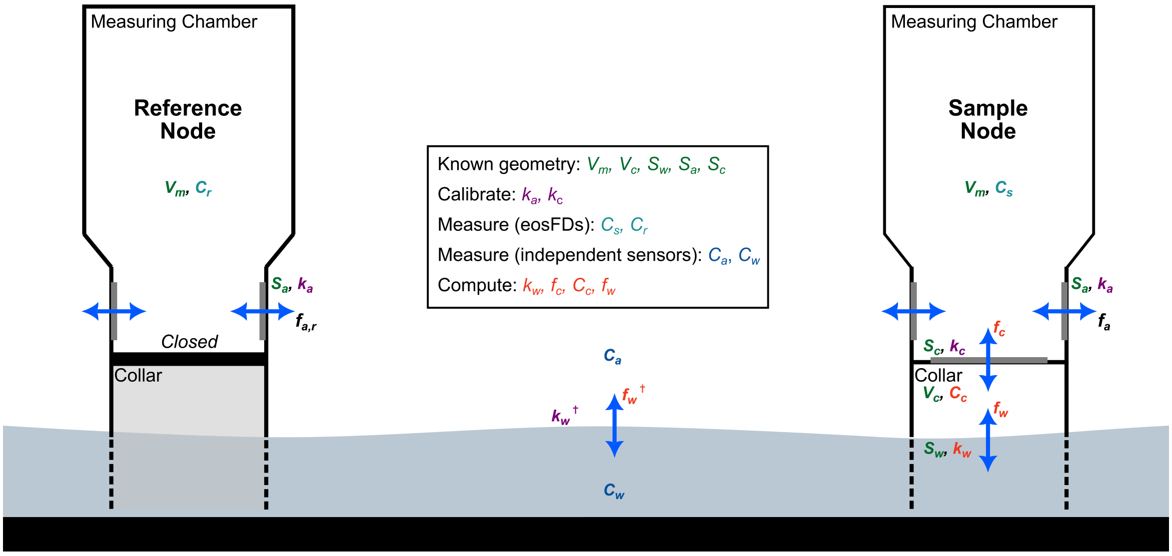

Figure 2: Diagram of an eosFD network (Sample and Reference Nodes), measuring over a water surface. Relevant parameters are shown, classified by whether they are known or must be calibrated, measured, or computed.

2.1 Non-Steady State Conditions

Assuming that a continuous dynamic equilibrium exists between the water and atmosphere, the air-water CO_2 flux rate beneath the Sample chamber (f_w) can be derived by considering the mass balance within the 1) Sample Node measuring chamber, 2) Reference Node measuring chamber, and 3) Sample Node collar.

Note: In this section, all concentrations and fluxes (C_a, C_s, f_a, f_c, etc.) are functions of time (i.e., $C_a(t), C_s(t)$, $f_a(t)$, $f_c(t)$, etc.), but the explicit t-dependence is omitted for readability. Time dependence is made explicit again when solving the ODE for $C_c(t)$ (Eq. 16).

2.1.1 Mass balance within the measuring chambers

The dynamic mass balance within the Sample Node's measuring chamber is as follows:

$$V_m \frac{\partial C_s}{\partial t} = S_c f_c - S_a f_a \tag{3}$$

Fick's Law gives the total CO_2 flux through the Sample Node side membranes:

$$f_a = k_a(C_s - C_a) \tag{4}$$

For the Reference Node, the bottom membrane membrane is sealed off, yielding the following dynamic mass balance:

$$V_m \frac{\partial C_r}{\partial t} = -S_a f_{a,r} \tag{5}$$

Fick's Law gives the total CO_2 flux through the Reference Node side membranes:

$$f_{a,r} = k_a(C_r - C_a) \tag{6}$$

Assuming C_a is uniform, the flux through the bottom membrane can be derived as follows.

For the Sample Node, substitute Eq. 2 into Eq. 1:

$$V_m \frac{\partial C_s}{\partial t} = S_c f_c - S_a k_a(C_s - C_a) \tag{7}$$

For the Reference Node, substitute Eq. 4 into Eq. 3:

$$V_m \frac{\partial C_r}{\partial t} = -S_a k_a(C_r - C_a) \tag{8}$$

Combine the two time derivatives (Eq. 5 and 6). This removes the dependence on C_a and any bias from atmospheric variability, as the Reference Node acts as a real-time correction for atmospheric exchange.

$$V_m \left(\frac{\partial C_s}{\partial t} - \frac{\partial C_r}{\partial t} \right) = S_c f_c - S_a k_a(C_s - C_r) \tag{9}$$

Finally, solve for f_c:

$$\boxed{f_c = \frac{V_m}{S_c} \left(\frac{\partial C_s}{\partial t} - \frac{\partial C_r}{\partial t} \right) + \frac{S_a}{S_c} k_a(C_s - C_r)} \tag{10}$$

This equation depends only on:

- geometry (S_a, S_c, V_m),
- the eosFD-measured concentrations (C_s, C_r),
- their time derivatives ($\frac{\partial C_s}{\partial t}$, $\frac{\partial C_r}{\partial t}$), and
- the side-membrane gas transfer velocity (k_a), obtained via calibration (or using the averaged Eosense value; see Table 4).

2.1.2 Mass balance within the collar

Similarly, the mass balance within the Sample Node collar is:

$$V_c \frac{\partial C_c}{\partial t} = S_w f_w - S_c f_c \tag{11}$$

The air–water CO_2 flux rate beneath the Sample chamber is given by the bulk flux equation,

$$f_w = k_w(C_w - C_c) \tag{12}$$

Fick's Law gives the CO_2 flux rate through the Sample Node bottom membrane:

$$f_c = k_c(C_c - C_s) \tag{13}$$

Substituting in Eqs. 10 and 11, the equation governing the CO_2 concentration in the collar (Eq. 9) can be rewritten as

$$\frac{\partial C_c}{\partial t} + \left(\frac{S_w k_w}{V_c} + \frac{S_c k_c}{V_c}\right) C_c = \frac{S_w k_w}{V_c} C_w + \frac{S_c k_c}{V_c} C_s \tag{14}$$

If we define the water-to-collar and collar-to-chamber transfer rate constants as $\kappa_w = S_w k_w / V_c$ and $\kappa_c = S_c k_c / V_c$ (units: s^{-1}), respectively, Eq. 12 becomes:

$$\frac{\partial C_c}{\partial t} + (\kappa_w + \kappa_c) C_c = \kappa_w C_w + \kappa_c C_s \tag{15}$$

The general solution of the first-order, non-homogeneous linear differential equation (Eq. 13) is given by the sum of the homogeneous and particular solutions:

$$C_c(t) = Ae^{-(\kappa_w + \kappa_c)t} + \frac{\kappa_w C_w + \kappa_c C_s}{\kappa_w + \kappa_c} \tag{16}$$

where A is a constant defined by initial conditions. If the initial concentration in the collar at $t = 0$ is $C_c(0) = C_a$, then A is given by

$$A = C_a - \frac{\kappa_w C_w^0 + \kappa_c C_s^0}{\kappa_w + \kappa_c} \tag{17}$$

The final solution for the concentration in the collar is then

$$\boxed{C_c(t) = \left(C_a - \frac{\kappa_w C_w^0 + \kappa_c C_s^0}{\kappa_w + \kappa_c}\right) e^{-(\kappa_w + \kappa_c)t} + \frac{\kappa_w C_w + \kappa_c C_s}{\kappa_w + \kappa_c}} \tag{18}$$

The time-dependent air–water CO_2 flux rate beneath the eosFD Sample chamber can then calculated using the bulk flux equation:

$$\boxed{f_w = k_w(C_w - C_c)} \tag{19}$$

Eqs. 18 and 19 depend on:

- known geometry (S_w, S_c, V_c),
- the eosFD Sample Node-measured CO_2 concentrations (C_s^0, C_s),
- the atmospheric CO_2 concentration (C_a), which must be measured with an external CO_2 sensor in air,
- the water CO_2 concentration (C_w), which must be measured with an external CO_2 sensor in water,
- the bottom-membrane gas transfer velocity (k_c), obtained via calibration, and
- the air–water gas transfer velocity beneath the chamber (k_w), which must be determined through analysis in the steady state (see next section).

2.2 Steady-State Conditions

In the steady state, we assume that

$$\frac{\partial C_s}{\partial t} = \frac{\partial C_r}{\partial t} = \frac{\partial C_c}{\partial t} = 0$$

From Eq. 8, it follows that $C_r = C_a$.

This assumption yields a simplified expression for the CO_2 concentration in the collar, C_c:

$$\begin{cases} (10) : f_c = \dfrac{S_a}{S_c}\, k_a\, (C_s - C_a) \\ (13) : f_c = k_c\, (C_c - C_s) \end{cases} \implies C_c = C_s + \frac{S_a k_a}{S_c k_c}\, (C_s - C_r) \tag{20}$$

which can then be used to calculate the gas transfer velocity beneath the collar:

$$\begin{cases} (11) : & S_w\, f_w &= S_c\, f_c \\ (13) : & f_c &= k_c\, (C_c - C_s) \\ (19) : & f_w &= k_w\, (C_w - C_c) \end{cases} \implies k_w = k_c\, \frac{S_c}{S_w}\, \frac{C_c - C_s}{C_w - C_c} \tag{21}$$

We can now derive a simple expression for the steady-state true air–water CO_2 flux, $f_w^\dagger$.

First, substitute Eq. 20 into Eq. 21. Given that $S_a = S_c = S_w$, this results in:

$$\boxed{k_w = \frac{k_a}{\dfrac{C_w - C_s}{C_s - C_r} - \dfrac{k_a}{k_c}}} \tag{22}$$

Assuming that $k_w^\dagger = k_w$, the bulk equation for the true air–water flux is:

$$f_w^\dagger = k_w (C_w - C_a) \tag{23}$$

Substituting in Eq. 22, we arrive at the steady-state true flux equation,

$$\boxed{f_w^\dagger = \frac{k_a (C_w - C_r)}{\dfrac{C_w - C_s}{C_s - C_r} - \dfrac{k_a}{k_c}}} \tag{24}$$

Notes:

- We can obtain k_a, k_c from calibration of the side and bottom membranes.
- The eosFD Sample Node provides C_s.
- We must measure C_w using an external water-side CO_2 sensor.
- The assumption that $C_r = C_a$ must be verified by comparing the eosFD Reference Node concentrations with that from an independent air-side CO_2 sensor.
- The derived expression for k_w relies on the uniformity and accuracy of the measured C_w; water mixing must be sufficient to establish a homogeneous concentration field.

2.3 Concentration-Based Calibration

Our goal is to ultimately measure the air–water flux (f_w). To do so, we must calibrate the chambers to determine the transfer velocities through the side membranes (k_a) and bottom membrane (k_c).

To calibrate for the side membranes, we can use both the Sample and Reference Nodes. The Reference Node's dynamic mass balance is given by Eq. 5. Similarly, if we cover the bottom membrane of the Sample Node, the mass balance within its measuring chamber follows from Eq. 3:

$$\frac{V_m}{S_a}\frac{\partial C_r}{\partial t} = -k_a(C_r - C_a) \tag{25a}$$

$$\frac{V_m}{S_a}\frac{\partial C_s}{\partial t} = -k_a(C_s - C_a) \tag{25b}$$

To calibrate for the bottom membrane, we can cover the Sample Node's side membranes; given that in this setup $C_c = C_a$, Eq. 3 becomes

$$\frac{V_m}{S_a}\frac{\partial C_s}{\partial t} = -k_c(C_s - C_a) \tag{26}$$

For a constant C_a, the general solution to the first-order linear ODEs in Eq. 25 and 26 is

$$C_m(t) = C_a + (C_m(0) - C_a)e^{-\gamma_i t}; \quad \gamma_i = \frac{S_i k_i}{V_m} \tag{27}$$

where the subscript m denotes either s or r, depending on whether the Sample or Reference chamber is used for calibration, and the subscript i denotes either a or c to refer to calibration of either the side (atmospheric) or bottom (collar) membranes. The exponent γ_i is a constant computed after fitting an exponential function to the sensor's time-response curve. Both S_i and V_m are provided by the manufacturer (Table 4), enabling k_i to be calculated.

In practice, electrical tape can be used to cover either the side or bottom membranes, and the eosFDs placed in a sealed environment in which the CO_2 concentration is increased from ambient conditions (e.g., 400–500 ppm) to a larger value (e.g., 900–1,000 ppm). Monitoring the exponential time response of the eosFDs to this step increase in CO_2 concentration over a sufficient time period enables an accurate fit to Eq. 27.

2.3.1 Comparison to Eosense Calibration

This calibration procedure differs in a key way from Eosense's. Their mass balance equation (Eq. 1) implicitly assumes that the flux through the bottom membrane (f_c) is the same as the flux directly from the sample surface (f_s)—i.e., they ignore the collar flux—and they calibrate for G, which is the gas transfer velocity of the *side membrane* (second term on the right in Eq. 1). They also assume that G is the same for the side and bottom membranes, which may not be true given that the path length from the side membrane to the NDIR sensor is not exactly the same as that from the bottom membrane to the sensor (Fig. 1).

This can be illustrated as follows. From Eqs. 3 and 5, the Sample and Reference mass balances are, respectively:

$$V_m\frac{\partial C_s}{\partial t} = S_c f_c - S_a f_a$$

$$\frac{\partial C_r}{\partial t} = -S_a f_{a,r}$$

Eosense assumes steady-state conditions: $\frac{\partial C_s}{\partial t} = \frac{\partial C_r}{\partial t} = 0$.

The Sample mass balance becomes

$$S_c f_c = S_a f_a$$

and similarly for the Reference mass balance:

$$S_a k_a (C_r - C_a) = 0$$
$$\implies C_r = C_a$$

Here, Eosense assumes that $f_c = f_s$, and prescribes this flux during their calibration. Given these assumptions, and since $S_a = S_c$, the Sample mass balance becomes:

$$f_s = f_a = k_a (C_s - C_r)$$

But in steady state, $C_a = C_r$; therefore,

$$f_s = k_a (C_s - C_r)$$

which is identical to Eq. 2, hence $G = k_a$; i.e., their calibration coefficient is equivalent to the side-membrane gas transfer velocity.

References

Nickerson, N., Lund, M., Simpson, G., Verfaillie, J., Ma, S. Y., Morrison, S., & McArthur, G. (2018, April 12). *Soil Respiration using Forced Diffusion: From the Tundra to the Savanna*. https://doi.org/10.13140/RG.2.2.18038.70720

Risk, D., Nickerson, N., Creelman, C., McArthur, G., & Owens, J. (2011). Forced Diffusion soil flux: A new technique for continuous monitoring of soil gas efflux. *Agricultural and Forest Meteorology*, *151*(12), 1622–1631. https://doi.org/10.1016/j.agrformet.2011.06.020